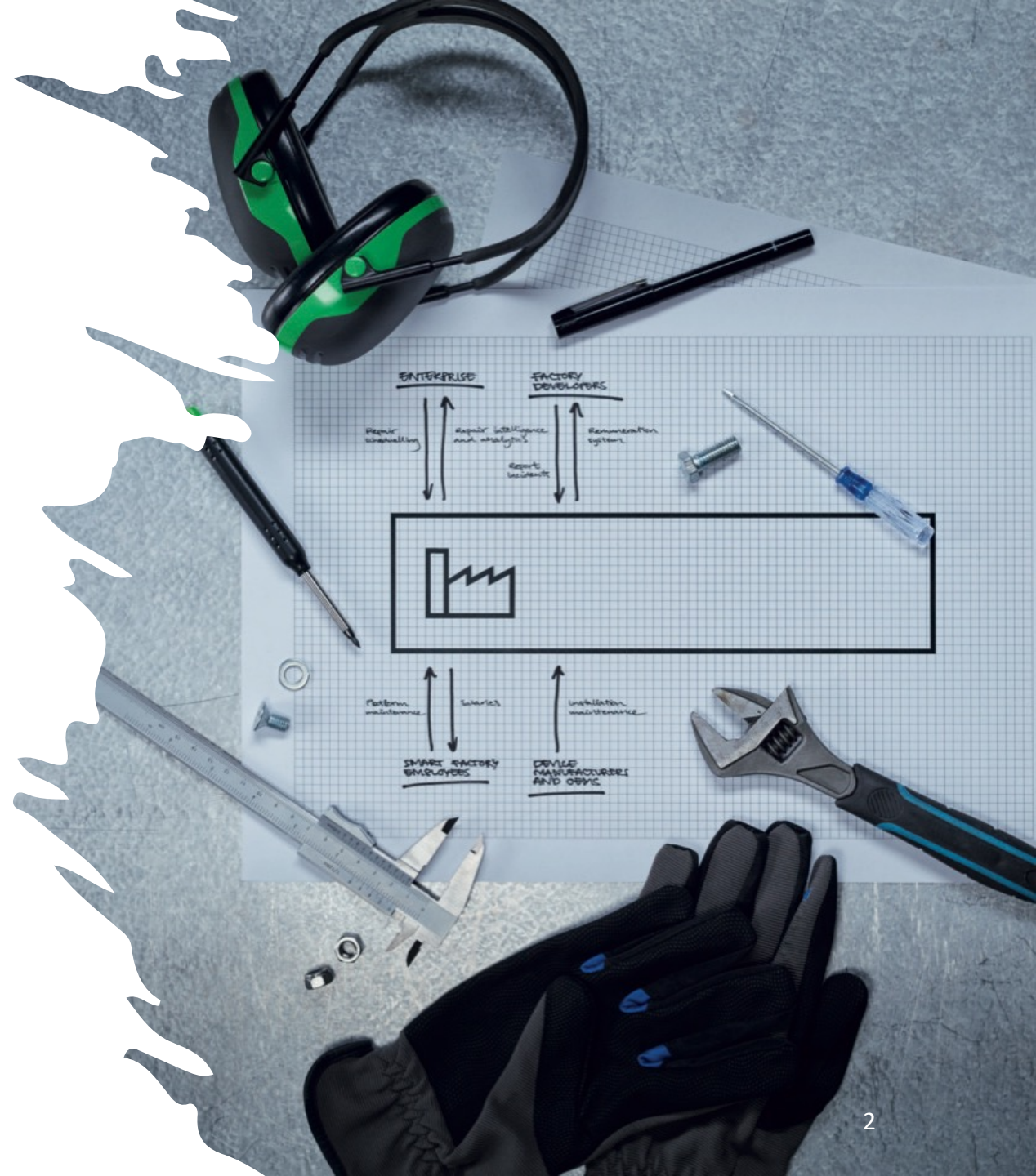


OMA Workshop Northbound API

<https://etherpad.openmobilealliance.org/p/2023-09-25-NBAPI.md>

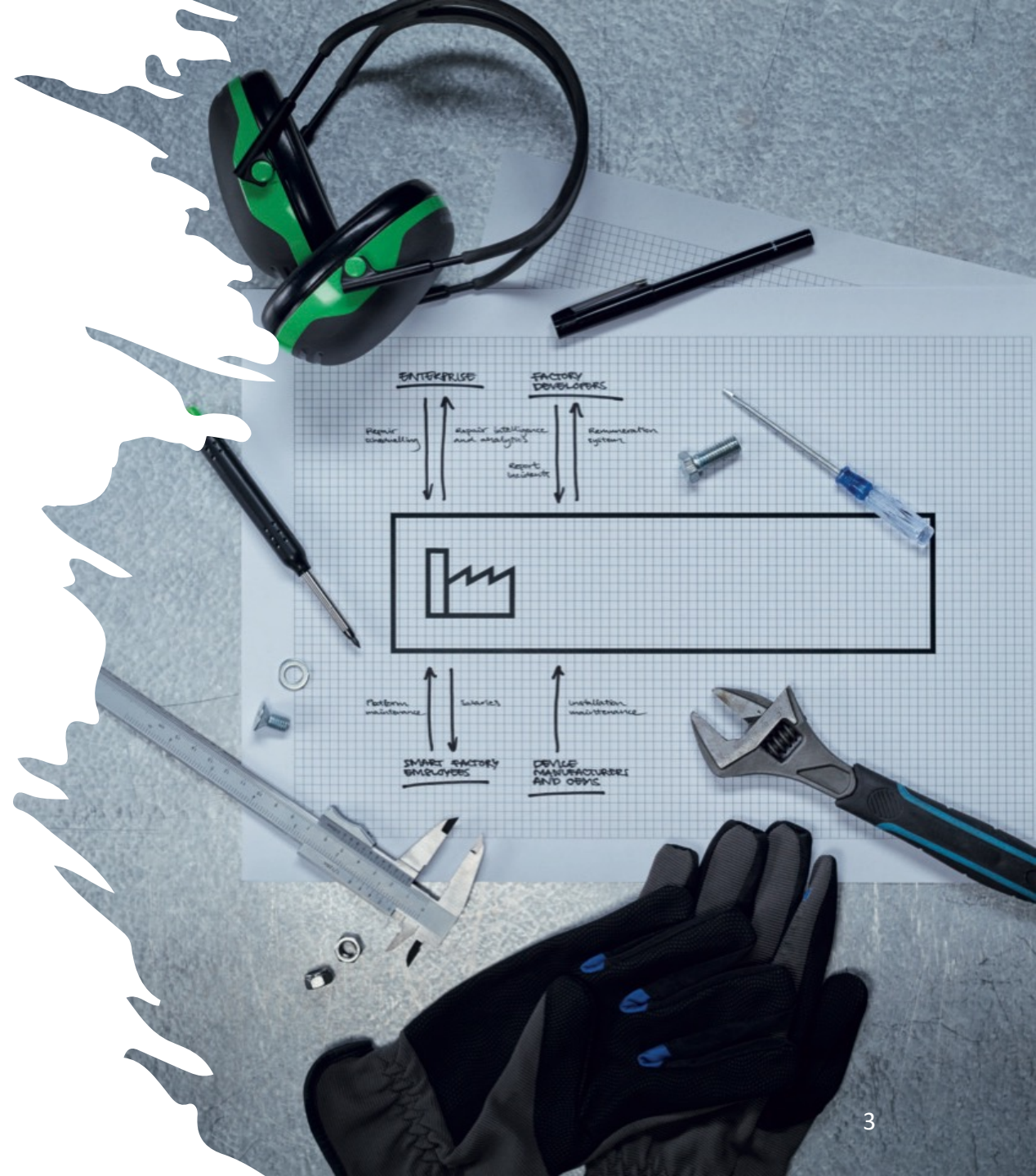
Agenda

- DMSO NB API
- Web APIs 101
- API usage scenarios
- Member views
- Open Discussion



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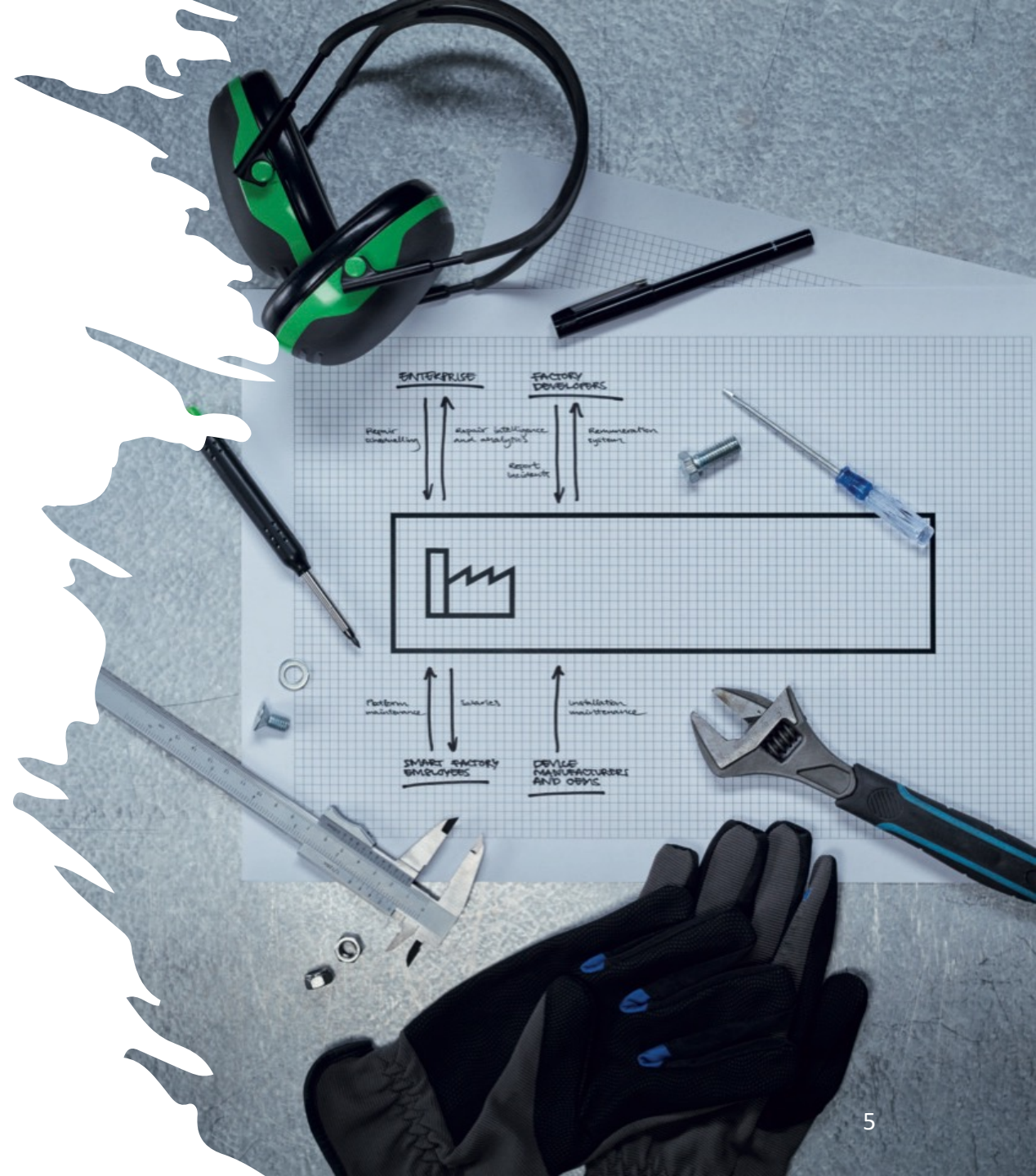


DMSO Northbound API

- DMSO has been chartered to build a standard “Northbound API” for LwM2M. An API would facilitate
 - ... system integration and interoperability.
 - ... simplification towards the developer community.
 - ... further adoption of the LwM2M Ecosystem.
- Our current task is to define the scope and the contents of that API.

Agenda

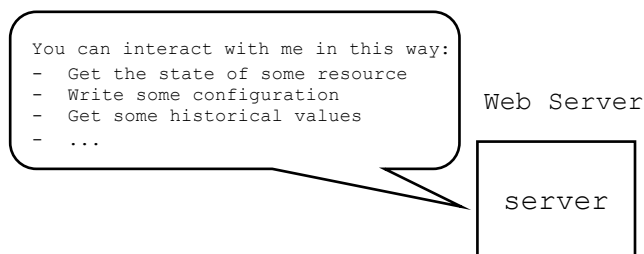
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Web API components

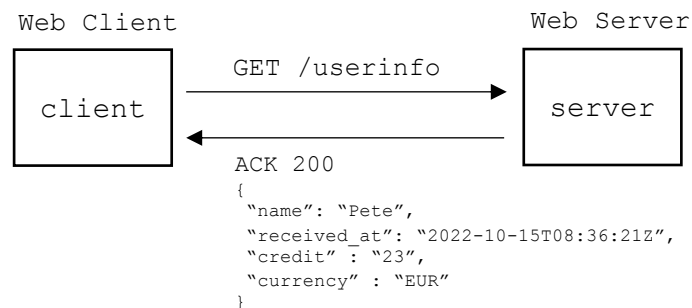
What is an API

- Application Programming Interface, an interface that a software program presents to other programs, humans or the World in the case of Web APIs.
- Born of the need to share information between data providers.
- APIs abstract software complexity.
- APIs are oriented to developers first (both internal and external) and as a product.



REST

- Representational State Transfer, an architectural style widely used on the Internet.
- Client-server architecture.
- REST is about state transfer and operating over *resources*.
- Resources are expressed as *URLs*, well-known *methods* with URLs inform the server of the actions to be done.



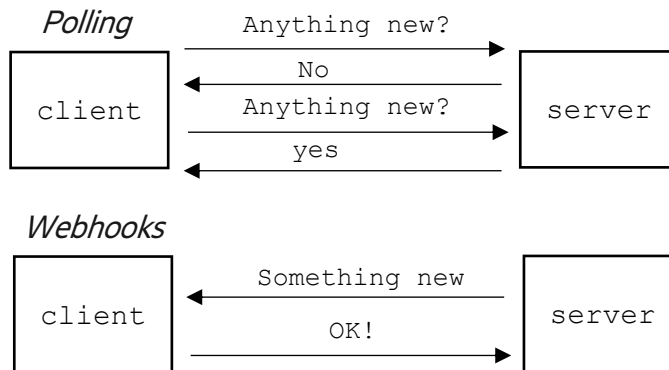
HTTP

- Typical protocol supporting REST is Hypertext Transfer Protocol (HTTP).
- GET https://company.org /users ?name=pete
 _/ _____/ _____/ _____/ _____/
method scheme authority path query
- Server hosting resources replies to HTTP requests (Request-Response).
 - Clients make HTTP requests and get back some JSON.
 - Many other relevant concepts like link attributes, media types, resource representations, query parameters, data semantics ... out of the scope of this presentation.
 - IoT protocols like CoAP are also RESTful.

Web API components

Real Time Data

- Responses can become old/stale.
- Polling is inefficient (only 1.5% return new data).
- A *webhook* allows to provide a URL to the server so that the “client” can receive notifications from the API.



Authentication & Authorization

- Need to authenticate parties using API (users, developers, apps).
- [RFC6749](#) (OAuth 2.0) allows limited access to a 3rd party to a HTTP service with access tokens.
- HTTP Authorization header on request

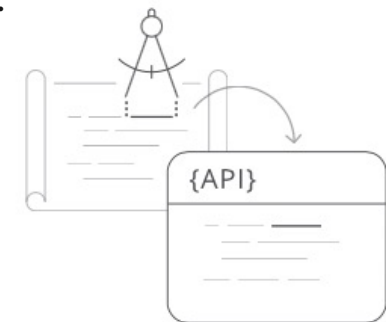
```
Authorization: Bearer xoxp-16501860-a24afj234
```

- User IDs and API Keys. Application ID and Application Private Keys.

```
# Your System API key
API_KEY='6f512343'
# Your System API secret
API_SECRET='kUO9ja5EXFnldkli'
# Your System ID
ID_NUMBER='35844299XXXX'
# Your System System Application ID
APP_ID='IDXYZ'
# Path to Application private key
PRIVATE_KEY='/Users/pete/private.key'
```

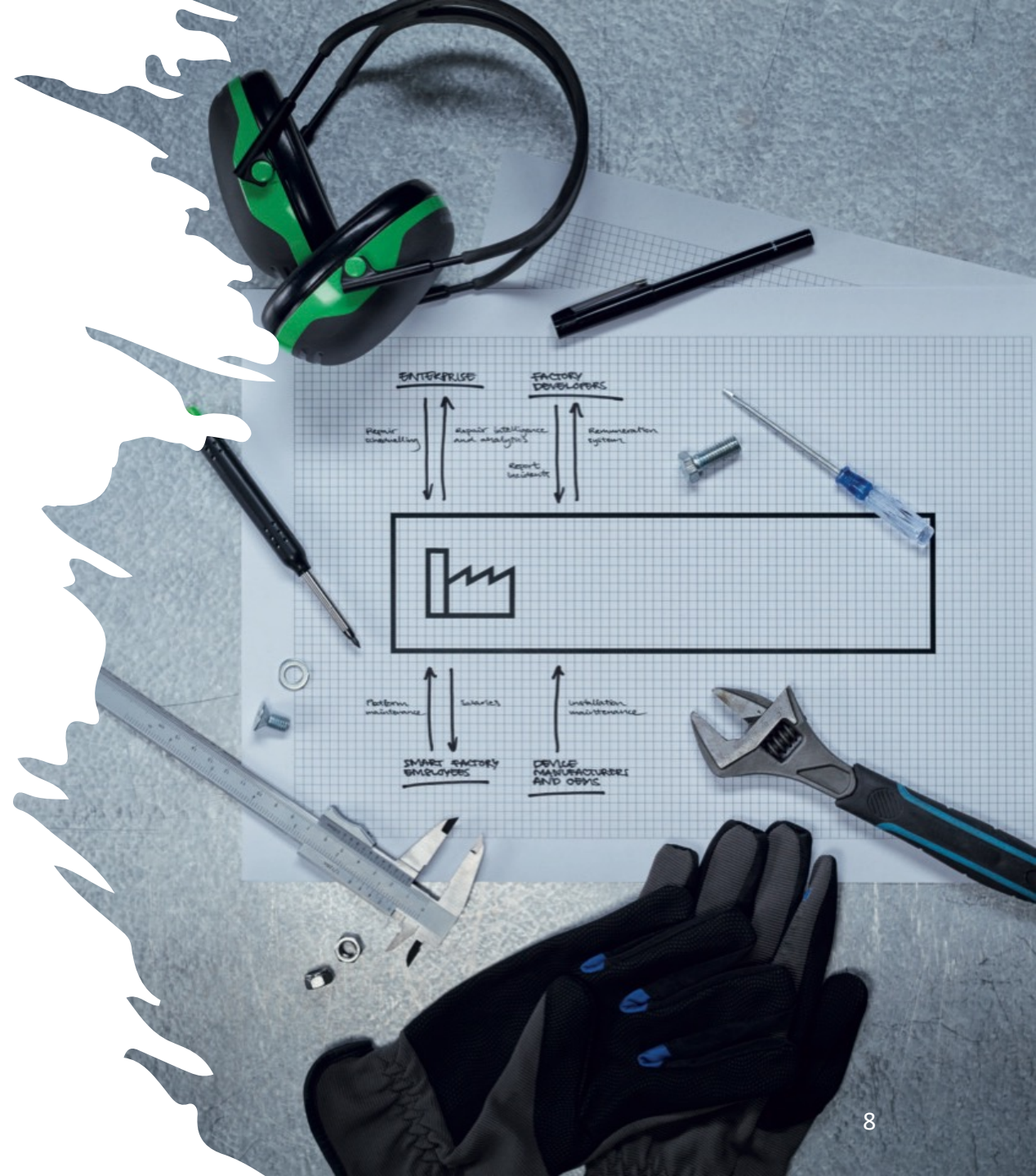
OpenAPI

- Common API Specification (OAS) format used on the web.
- Provides a human-readable specification of the interface for developers that need to use it.
- API is defined as a [YAML file](#), with methods to be used, resource paths, response codes, errors, etc.
- Swagger is the well-known editor with visualization, compliance, etc. Lots of tools available for working with APIs.



Agenda

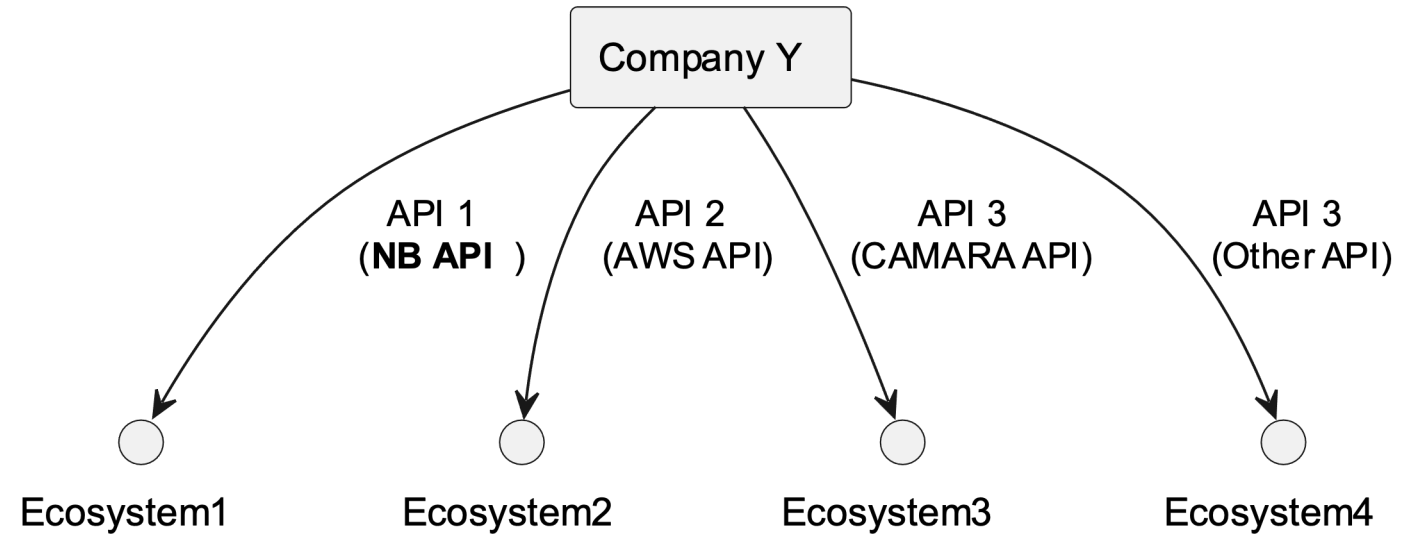
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API Usage Scenarios

The Ecosystem Aggregator

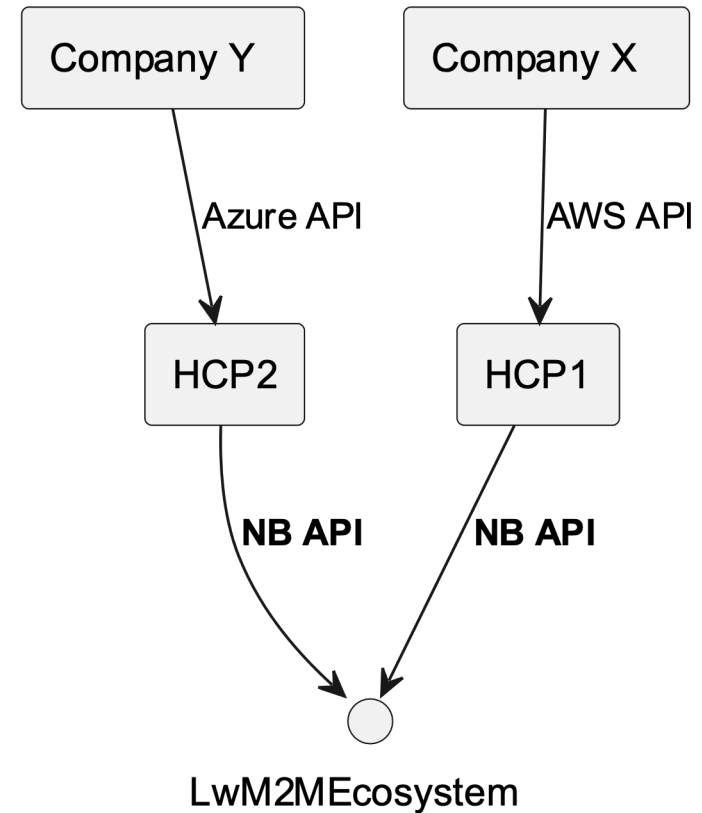
- The company intends to add a LwM2M API, which aligns with existing APIs to some degree in terms of resources, semantics, device ids...
- The company interacts with various ecosystems through these APIs.
- Device Management is one of the ecosystems and it needs a general-purpose NB LwM2M API that is modular and flexible to further abstract it to its business needs.



API Usage Scenarios

The HCP Integrator

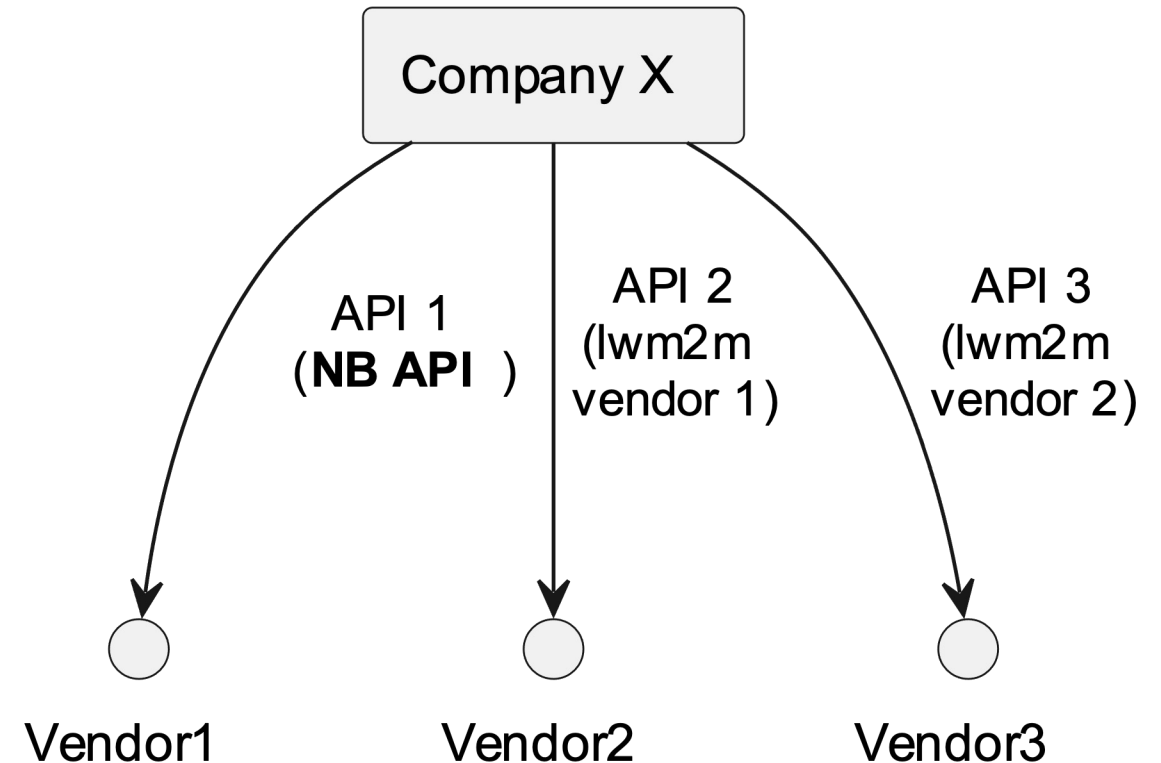
- The Hyperscale Cloud Provider (HCP) is a vendor that interacts with a device management ecosystem and exposes services over an API.
- The HCP tries to make integration with HCP platforms easier
- Company X and Y use different Hyperscalers that under the hood use a common API towards a LwM2M Ecosystem.
- The NB API aligns with existing APIs from HCP to some degree in terms of resources, semantics, device ids...
- Device Management is one of the ecosystems and it needs a general-purpose LwM2M API that is modular and flexible to further abstract it to its business needs.



API Usage Scenarios

The System Integrator

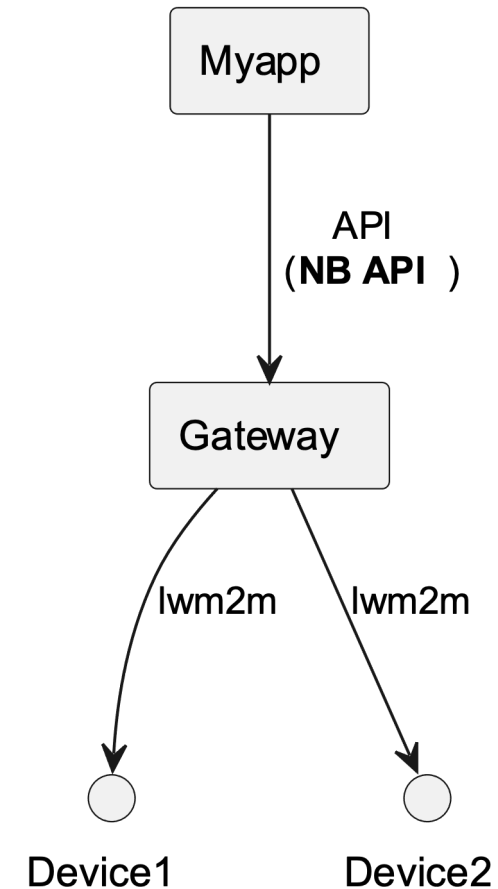
- The System Integrator is a company with multiple similar APIs for different device types.
- The company uses various device management vendors who implement their APIs on top of LwM2M.
- The company aims to have a single API for managing these vendors, ideally a standard that LwM2M server vendors implement.
- The company interacts with different LwM2M vendors via these APIs.
- This case is similar to the Ecosystem Integrator but operates entirely within the LwM2M ecosystem.
- The System Integrator enables seamless "enterprise business integration" in a vendor-agnostic manner.



API Usage Scenarios

The Application Developer via GW

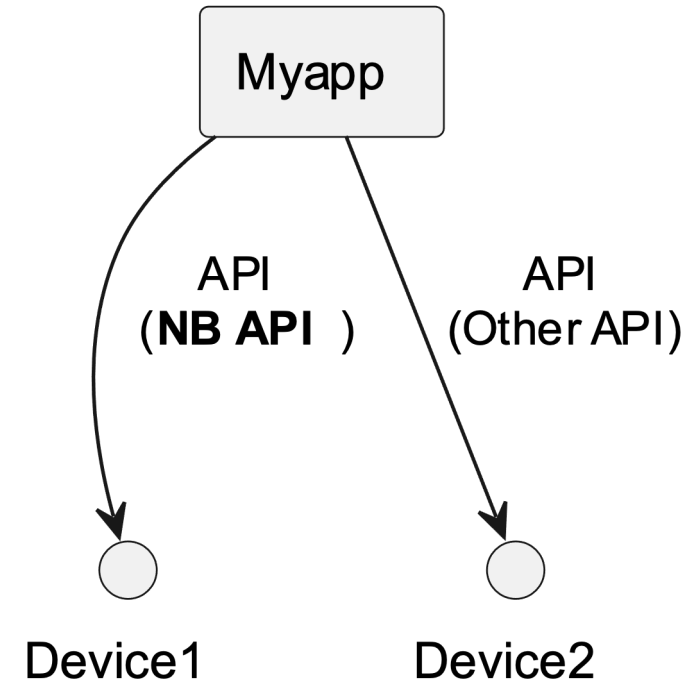
- The Application Developer is a company or user that aims to implement an application for managing their devices.
- A device vendor utilizes this API to expose devices under a LwM2M server, bypassing the complexity.
- The Application Developer interacts directly and locally via a Gateway.
- This represents a more localized scenario, specifically for gateway or router providers.
- The Gateway vendor can offer this API to abstract the LwM2M complexity from its developer community.



API Usage Scenarios

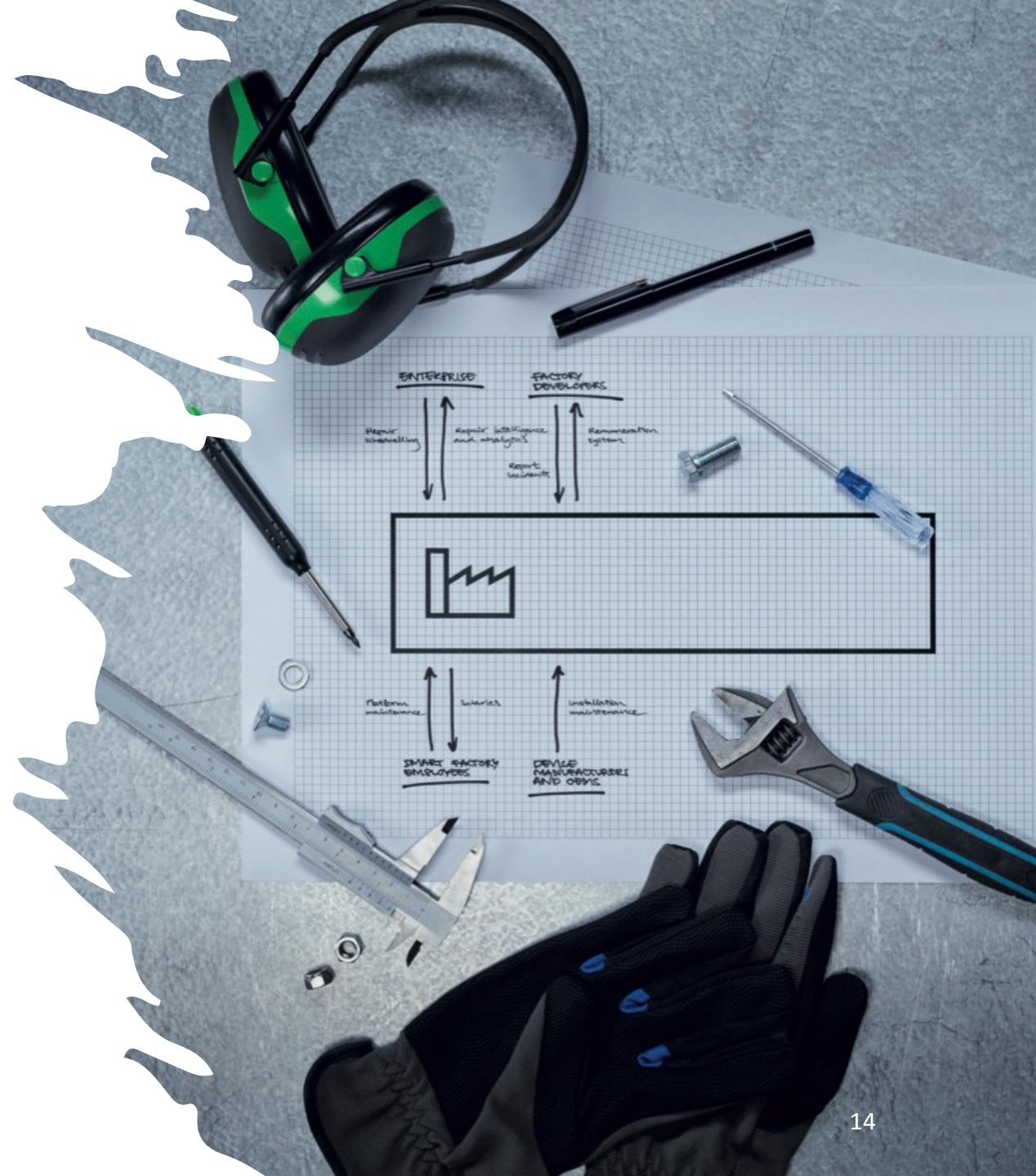
The Application Developer directly

- The Direct Application Developer is similar to the Application Developer but interacts directly with the LwM2M endpoints.
- The Direct Application Developer uses the API to expose devices managed by a LwM2M server and direct local access to them via this API.
- Unlike the Application Developer, this role does not rely on a Gateway for interaction but communicates directly with the LwM2M endpoints.
- LwM2M endpoints also implement the NB API locally - or over the Internet - in addition to their LwM2M interfaces.



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OneM2M APIs

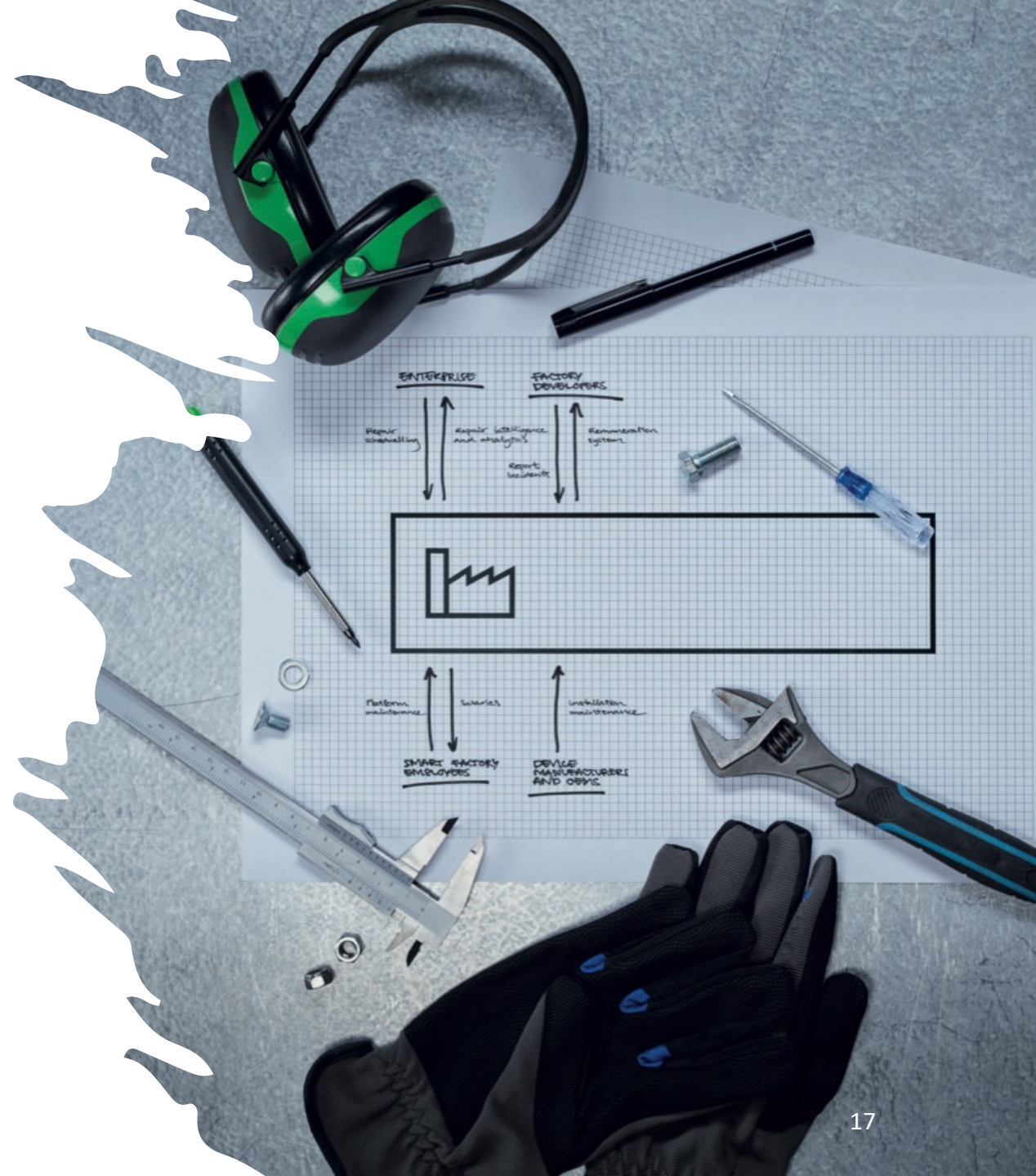
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CAMERA APIs

- Notes here

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Open Discussion

- Which are the primary use cases for our LwM2M northbound API?
- What existing APIs and models should we consider as references or benchmarks?
- What kind of abstractions (if at all), do we imagine would be beneficial in the NB APIs?
 - For example: just a device-side to server-side mapping?
- How can we in DMSO generate more interest on this topic from potential API consumers?
- How could we leverage the OMA Utilities (Outreach & Interoperability) channel?
- How could we leverage work happening in other SDOs, for example, CAMARA, TMF, etc.?
- Do we see any risks/ roadblocks in pursuing the work?